Corrected Manuscript

Revised trial report with updated references

iPhone-friendly edition — 2026-07-26

Contents

Corrected Manuscript

Revised manuscript draft

> **Editorial status:** This is a scientifically conservative reconstruction based only on the aggregate information in the supplied PDF. Bracketed fields require author verification. Confirmatory statistical results must be inserted after reanalysis of participant-level data. This draft must not be submitted until the ethics, registration, randomization, and data-audit items are resolved.

Title

Supraorbital versus infraorbital external trigeminal nerve stimulation for migraine prevention: a three-arm assessor-blinded randomized sham-controlled trial

Use “randomized” only if the allocation records verify random sequence generation and implementation.

Abstract

Background

External trigeminal nerve stimulation is a non-invasive option for migraine prevention, but evidence directly comparing stimulation sites is limited. We assessed supraorbital and infraorbital transcutaneous electrical nerve stimulation (TENS) against an inert sham condition.

Methods

In this single-center, three-arm trial, adults with migraine were allocated to supraorbital TENS (n=28), infraorbital TENS (n=27), or supraorbital sham stimulation (n=28). Active stimulation was administered in clinic at 100 Hz and 200 μs for 30 min per session, six sessions weekly for four weeks. The sham device delivered no current. The prespecified primary outcome was [verify against the dated registry and protocol] monthly migraine attack frequency at three months. Secondary outcomes were pain intensity, MIDAS, HIT-6, acute oral analgesic-use frequency, and MPQ-5 score. Outcome assessment and analysis were masked to allocation; successful participant masking was not established. Primary comparisons should be reanalyzed using a baseline-adjusted model with multiplicity control.

Results

All 83 allocated participants were reported to have completed three-month follow-up. At three months, the unadjusted mean monthly attack frequencies were 6.59 (SD 2.70), 6.16 (SD 2.48), and 12.94 (SD 5.01) in the supraorbital, infraorbital, and sham groups, respectively. The corresponding unadjusted endpoint mean differences versus sham were −6.35 attacks/month (approximate Welch 95% CI −8.52 to −4.18) and −6.78 attacks/month (−8.92 to −4.64). These aggregate-data contrasts are descriptive pending baseline-adjusted participant-level reanalysis. The supraorbital–infraorbital contrast was inconclusive (0.43 attacks/month, −0.97 to 1.83). Mild local sensations were commonly reported in both active groups; no serious adverse event was detected in this small, short trial.

Conclusions

The reported aggregate data show a preliminary efficacy signal for both active protocols relative to an inert sham in patients able to attend an intensive clinic schedule. Uncertain allocation details, likely compromised participant masking, a site-mismatched sham, outcome-measure limitations, and insufficient power for the active-site comparison preclude confirmatory conclusions about efficacy or equivalence. Verification of prospective registration and randomization, followed by prespecified reanalysis of participant-level data, is required.

Trial registration: IRCT [identifier; URL; registration date].

Ethics approval: [committee; approval identifier; approval date].

First participant enrolled: [date].

Keywords: migraine; external trigeminal nerve stimulation; transcutaneous electrical nerve stimulation; supraorbital nerve; infraorbital nerve; randomized trial

---

1. Introduction

Migraine is a leading cause of years lived with disability worldwide and is characterized by recurrent headache and associated sensory and autonomic symptoms [1,2,34]. The GBD 2023 analysis confirms that headache disorders remain highly prevalent and that migraine accounts for most headache-attributed years lived with disability [53]. An updated global synthesis estimated the prevalence of active migraine at 14.0% and headache on at least 15 days per month at 4.6% [41]. Migraine imposes substantial individual, economic, and health-system burdens [42]. Current models describe altered processing across cortical, subcortical, brainstem, and trigeminovascular networks [43]. Preventive pharmacotherapy can reduce migraine burden, but contraindications, inadequate response, adverse effects, and medication overuse motivate the evaluation of non-pharmacological options [10,50].

Non-invasive neuromodulation can alter nociceptive signaling without an implanted device. External stimulation of the supraorbital and supratrochlear branches of the trigeminal nerve has been evaluated for migraine prevention, including in a small sham-controlled trial and systematic reviews [15,26,44,46,55]. The certainty of evidence remains limited, and current recommendations are device-specific and conditional [47]. Efficacy estimates cannot be generalized across devices, electrode geometries, targets, waveforms, or dose schedules. Direct evidence for surface infraorbital stimulation in migraine is lacking; one small uncontrolled study evaluated combined supraorbital and infraorbital lidocaine blocks and could not isolate the infraorbital contribution [23,54].

We therefore evaluated two facial TENS placements—supraorbital and infraorbital—against an inert supraorbital sham condition. The primary objective was to estimate each active treatment's effect on monthly migraine attack frequency at three months relative to sham. The comparison between active sites was exploratory because the study was not designed for equivalence or non-inferiority.

---

2. Methods

2.1 Study design, reporting, and oversight

This single-center, three-arm, parallel-group trial was conducted at the outpatient Physical Medicine and Rehabilitation clinics of Isfahan University of Medical Sciences, Iran, between [first recruitment date] and [final follow-up date].

The protocol was approved by [full committee name] on [date] (approval [identifier]) and registered at the Iranian Registry of Clinical Trials on [date] (IRCT [identifier]; [URL]) before enrollment of the first participant on [date]. Written informed consent was obtained from all participants. Any discrepancy between the original registry/protocol and this report is listed in Supplementary Table S1 with its date, rationale, and whether it occurred before access to outcome data.

Reporting follows CONSORT 2025, CONSORT Harms 2022, the CONSORT extension for non-pharmacological interventions, and TIDieR [48,49]. The dated protocol and statistical analysis plan are available at [repository/URL].

> If registration occurred after enrollment, replace the prospective-registration sentence with the factual chronology and explain the delay.

2.2 Participants

Adults aged 18–65 years were eligible if they met ICHD-3 criteria for migraine with or without aura and had at least two migraine attacks monthly during a prospective [28-day] baseline diary period [2]. Authors must specify:

- the required number of migraine and headache days;

- episodic or chronic migraine status;
- the algorithm distinguishing a new attack from recurrence;
- whether a pain-free interval of 24 or 48 hours was used and why;
- minimum diary completion;
- permitted acute medications and their effect on classification.

Exclusion criteria were medication-overuse headache; another active headache disorder; pregnancy or lactation; [device-specific contraindications]; severe psychiatric or systemic illness; substance use disorder; specified preventive procedures within [class-specific washout periods]; and current preventive medication unless a stable regimen was explicitly permitted by the protocol.

“Treatment non-compliance” must not be listed as a pre-randomization exclusion unless it was operationalized prospectively. Post-randomization nonadherence should be retained in the intention-to-treat population.

2.3 Recruitment and applicability

Of 137 screened individuals, 54 did not proceed to randomization because they were unable or unwilling to commit to 24 clinic visits over four weeks. Eighty-three participants entered the allocation process. The high pre-randomization exclusion fraction restricts applicability to patients who can attend this intensive schedule. Because the 54 individuals were not randomized and had no study outcomes, no outcomes were imputed for them.

Report available age, sex, migraine subtype, baseline frequency, and reasons for nonparticipation for screened nonrandomized individuals when permitted by ethics approval.

2.4 Randomization and allocation concealment

Replace this section using the original allocation records:

> An independent [role] generated the allocation sequence using [software/version and exact algorithm] with a 1:1:1 ratio and [actual block sizes/stratification factors]. [Role], who had no contact with participants, prepared sequentially numbered, opaque, tamper-evident, sealed envelopes. After eligibility and consent were completed, [role] opened the next envelope in sequence. Treatment providers could not be masked because they operated the device.

A fixed block size of four cannot by itself produce balanced 1:1:1 permuted blocks. Do not retain that statement unless the exact unequal-block rotation algorithm and its implementation can be documented.

2.5 Masking

Participants were intended to be unaware of treatment allocation. However, active stimulation was titrated to a strong sensory level, whereas sham stimulation delivered no current. Successful participant masking therefore cannot be assumed. Outcome assessment and statistical analysis were conducted by an investigator unaware of allocation [describe safeguards].

Report the masking assessment as a full cross-tabulation of assigned group by guessed group, including “do not know,” the timing of assessment, and reasons for each guess. Analyze it with an appropriate blinding index and confidence interval. Do not infer successful masking from a nonsignificant difference in correct-guess proportions.

2.6 Interventions

All intervention details below require confirmation from treatment logs and device documentation.

Supraorbital TENS

Bilateral stimulation was administered over the supraorbital foramina/notches with [device manufacturer, exact model, regulatory status]. Sessions lasted 30 min and occurred six times weekly for four weeks (24 planned sessions). Frequency was 100 Hz and pulse duration was 200 μs. Current was individually increased to a strong but tolerable sensation; the actual delivered intensity had a median of [value] mA (IQR [value]) and range [value].

Infraorbital TENS

The same device, schedule, and nominal electrical parameters were used, with electrodes centered over the infraorbital foramina. Authors must describe precautions used for facial and ocular proximity.

Sham stimulation

Electrodes were placed over the supraorbital region using the same visible session procedure, but no current was delivered. This inert sham did not reproduce treatment sensation and did not site-match the infraorbital intervention.

For all groups, report:

- waveform and polarity;
- electrode dimensions, material, spacing, and montage;
- anatomical localization method;
- skin preparation and impedance threshold;
- ramp-up/ramp-down;
- calibration and output verification;
- provider training;
- protocol-fidelity checks;
- deviations and actual attendance per arm.

2.7 Outcomes

Primary outcome

The primary outcome was [verify from original protocol and registry] monthly migraine attack frequency during [exact diary window] at three months. Define the unit, denominator, standardization for incomplete/unequal diary windows, and rules for recurrence.

If monthly migraine days were collected, they should be reported because current preventive-trial guidance recommends change in monthly migraine days from a prospective 4–8-week diary baseline [52]. Report ≥50% responder rates only if prespecified or clearly label them exploratory.

Secondary outcomes

- 1. Pain intensity measured with [10-cm VAS or 0–10 NRS], anchors [specify], referring to [average/worst/typical pain] over [recall window].
- 2. MIDAS total score, using the validated Persian version [51] and its standard three-month recall period. The one-month assessment overlaps the pretreatment period and should not be interpreted as an isolated one-month treatment effect.
- 3. HIT-6 total score, using the correctly cited validated Persian version [38].
- 4. Number of acute oral analgesic-use occasions during [exact 28-day window], with medication classes and counting rules defined.
- 5. MPQ-5 score, reported as exploratory unless a validated Persian version and longitudinal responsiveness can be documented. Provide score range, item scoring, recall period, and interpretation.

Adverse events should be elicited with a standardized checklist plus open-ended inquiry, and coded by severity, seriousness, expectedness, relatedness, onset, duration, and action taken.

2.8 Concomitant treatment

Define whether preventive medications were prohibited, continued at a stable dose, or allowed as rescue. List class-specific washout periods. A universal two-week washout is not adequate for all long-acting preventive treatments. Acute medication use must be incorporated into the estimand because it may affect pain reporting and is also a secondary outcome.

2.9 Sample size

Report the original, dated protocol calculation exactly, including its endpoint, anticipated effect, variance/event rate, alpha, power, number of primary comparisons, test, and attrition allowance.

If the original calculation was based on responder rate, state that calculation and disclose that the continuous/count outcome analysis differs from the original design. A retrospective calculation based on observed or selected assumptions must not be presented as prospective justification. Remove observed post-hoc power.

For reference only, a simple two-sample normal approximation with SD 4.5, difference 3, two-sided α=.05, and 80% power requires approximately 36 evaluable participants per group. Allowing 20% attrition requires approximately 45 randomized participants per group after rounding conservatively. This calculation does not account for multiple active–sham comparisons and is not a replacement for the original design.

2.10 Statistical analysis

The final analysis must be rerun from participant-level data under a dated analysis plan prepared without reference to treatment-coded results.

Recommended primary model:

- baseline-adjusted negative-binomial regression for monthly attack count, with log observation time as an offset when diary windows differ;
- fixed effects for treatment and baseline attack frequency;
- two prespecified active-versus-sham contrasts;
- Holm or Dunnett control of the primary family-wise error rate;
- adjusted rate ratios and adjusted absolute marginal mean differences with 95% CIs.

If model fit is inadequate, use a justified alternative and provide diagnostics. A baseline-adjusted ANCOVA with robust standard errors may be a sensitivity analysis if the outcome behaves approximately continuously.

Recommended secondary analysis:

- baseline-adjusted ANCOVA or suitable generalized models at three months;
- a longitudinal mixed model/GEE with group, time, and group×time interaction as supportive analysis;
- multiplicity control across the prespecified secondary family;
- exact analysis population, missing-data strategy, and intercurrent-event strategy.

The active-site contrast is exploratory and must not be interpreted as equivalence. Report medians and IQRs if rank-based analyses are retained, and use effect estimates compatible with those tests.

Analyses that add nonrandomized screened individuals to sham are prohibited because they destroy the randomized comparison and do not identify transportability.

---

3. Results

3.1 Participant flow

Report exact dates and include a readable CONSORT diagram:

- assessed for eligibility: 137;

- excluded before allocation: 54, with mutually exclusive reasons;
- entered allocation: 83;
- received allocated intervention: [verify per arm];
- completed each time point: [verify per arm];
- included in each analysis: [verify per arm].

The manuscript reports no post-allocation loss to follow-up. Support this unusual result with attendance and diary-completion logs rather than attributing it to tolerability.

3.2 Baseline characteristics

Do not test baseline “homogeneity.” Present descriptive clinical characteristics by allocated group:

Characteristic	Supraorbital (n=28)	Infraorbital (n=27)	Sham (n=28)
Age, years, mean (SD)	35.8 (7.6)	36.4 (6.9)	35.1 (7.2)
Female, n (%)	21 (75.0)	20 (74.1)	22 (78.6)
Migraine with aura, n (%)	[data]	[data]	[data]
Chronic migraine, n (%)	[data]	[data]	[data]
Disease duration, years	[data]	[data]	[data]
Headache days/28 days	[data]	[data]	[data]
Migraine days/28 days	[data]	[data]	[data]
Migraine attacks/28 days	14.18 (3.95)	15.13 (4.51)	12.32 (4.99)
VAS/NRS	7.30 (1.29)	7.07 (1.47)	7.10 (1.11)
MIDAS	20.02 (5.22)	17.94 (6.30)	19.36 (4.05)
HIT-6	62.58 (4.78)	64.22 (5.93)	61.11 (3.74)
Analgesic-use occasions	14.62 (4.35)	14.72 (3.50)	14.36 (4.27)
MPQ-5	26.43 (6.49)	26.52 (6.23)	25.42 (6.40)
Previous preventive failures	[data]	[data]	[data]
Acute medication classes	[data]	[data]	[data]

Check all values against participant-level data. The age and sex p-values in the supplied manuscript are not reproducible from the aggregate table and should be removed.

3.3 Primary outcome

The observed aggregate values were:

Group	Baseline mean (SD)	1 month mean (SD)	3 months mean (SD)	Unadjusted mean change
Supraorbital	14.18 (3.95)	7.68 (2.84)	6.59 (2.70)	−7.59
Infraorbital	15.13 (4.51)	9.19 (4.03)	6.16 (2.48)	−8.97

Group	Baseline mean (SD)	1 month mean (SD)	3 months mean (SD)	Unadjusted mean change
Sham	12.32 (4.99)	14.27 (4.15)	12.94 (5.01)	+0.62

At three months, the unadjusted endpoint differences were:

Contrast	Mean difference (approximate Welch 95% CI)
Supraorbital − sham	−6.35 (−8.52 to −4.18)
Infraorbital − sham	−6.78 (−8.92 to −4.64)
Supraorbital − infraorbital	0.43 (−0.97 to 1.83)

These values are descriptive and do not account for baseline differences. Replace the following template after participant-level reanalysis:

> In the prespecified baseline-adjusted model, the estimated [rate ratio/absolute marginal difference] for supraorbital TENS versus sham was [estimate] (multiplicity-adjusted 95% CI [x to y], p=[value]), and for infraorbital TENS versus sham was [estimate] ([x to y], p=[value]). The exploratory supraorbital–infraorbital estimate was [estimate] ([x to y]); this trial did not test equivalence or non-inferiority.

Do not use separate within-group Friedman p-values as evidence of treatment efficacy.

3.4 Secondary outcomes

Aggregate three-month outcomes and corrected descriptive Welch intervals:

Outcome	Supra mean (SD)	Infra mean (SD)	Sham mean (SD)	Supra−sham MD (95% CI)	Infra−sham MD (95% CI)	Supra−infra MD (95% CI)
Pain intensity	3.92 (0.97)	3.90 (1.33)	6.53 (1.02)	−2.61 (−3.14, −2.08)	−2.63 (−3.27, −1.99)	0.02 (−0.61, 0.65)
MIDAS	9.76 (3.12)	10.30 (2.07)	15.96 (5.39)	−6.20 (−8.57, −3.83)	−5.66 (−7.88, −3.44)	−0.54 (−1.97, 0.89)
HIT-6	49.48 (4.84)	51.40 (3.01)	62.18 (4.13)	−12.70 (−15.11, −10.29)	−10.78 (−12.73, −8.83)	−1.92 (−4.10, 0.26)
Analgesic use	5.02 (2.64)	5.65 (2.55)	12.06 (4.25)	−7.04 (−8.94, −5.14)	−6.41 (−8.31, −4.51)	−0.63 (−2.03, 0.77)
MPQ-5	16.77 (5.08)	14.87 (5.34)	24.16 (4.23)	−7.39 (−9.90, −4.88)	−9.29 (−11.91, −6.67)	1.90 (−0.92, 4.72)

These normal-theory endpoint intervals do not replace baseline-adjusted models and should not be paired with Mann–Whitney p-values as if they estimated the same parameter. Insert adjusted estimates and multiplicity-controlled results after reanalysis.

Do not assign a HIT-6 category to a group mean; report individual category transitions if clinically relevant.

3.5 Treatment exposure and adherence

Report planned and received sessions by group:

Exposure measure	Supraorbital	Infraorbital	Sham
Participants receiving ≥1 session	[data]	[data]	[data]
Sessions received, median (IQR)	[data]	[data]	[data]
Completed all 24 sessions, n (%)	[data]	[data]	[data]
Delivered current, mA, median (IQR)	[data]	[data]	0
Protocol deviations, n	[data]	[data]	[data]

3.6 Masking

Replace the summary percentages with the complete assigned-group by guessed-group table and a valid blinding index. State:

> Because sham delivered no current and active treatment commonly produced paresthesia, participant masking was not considered established.

3.7 Adverse events

The supplied report identified itching in 9/28 participants in the supraorbital group and 8/27 in the infraorbital group, and tingling/paresthesia in 12/28 and 11/27, respectively. Verify whether participants could report more than one event and provide sham-arm local-event counts.

No serious adverse event was detected. This observation does not establish safety; the approximate rule-of-three upper 95% bound for an unobserved serious-event proportion is 3.6% across 83 participants and approximately 11% in each active arm.

Report all events in a CONSORT harms table with severity, duration, relatedness, and withdrawals.

---

4. Discussion

4.1 Principal findings

The aggregate data showed lower reported migraine attack frequency at three months in both active-stimulation groups than in the inert sham group. Similar descriptive patterns occurred for pain intensity, disability, headache impact, and medication-related measures. These findings represent a preliminary signal rather than confirmatory evidence because the primary analysis requires baseline-adjusted participant-level reanalysis and successful participant masking was not established.

The active-site contrast was imprecise and the study was not designed for equivalence or non-inferiority. Therefore, the absence of a statistically significant difference must not be interpreted as equivalence, interchangeability, or evidence that site selection can be based solely on preference.

4.2 Relation to previous evidence

One small sham-controlled trial and subsequent evidence syntheses support a possible preventive effect of device-specific external trigeminal stimulation protocols [15,26,44,46,55]. However, pooled estimates are based on few heterogeneous studies, and one later synthesis found no significant benefit for several eTNS monotherapy outcomes [30]. Current IHS guidance makes only conditional recommendations for named devices and does not endorse generic or infraorbital TENS [47]. These results cannot be transferred directly to the Novin device and protocol used here because waveform, electrode configuration, delivered dose, target, and treatment schedule differ.

A 2026 open-label active-controlled trial directly compared supraorbital tSNS with low-dose topiramate [56]. The tSNS arm had fewer adverse events but a lower responder rate (20.6% vs 33.9%) and did not meet the prespecified non-inferiority criterion. This result provides contemporary comparative context but does not validate the present protocol because the device, comparator, masking, and stimulation schedule differed.

Direct evidence supporting surface infraorbital TENS for migraine is lacking. The combined nerve-block study [23] is indirect, and a contemporary peripheral-nerve-stimulation review likewise identifies stronger evidence for supraorbital and occipital targets than for infraorbital stimulation in migraine [54]. Reference 24 concerns implanted occipital plus supraorbital—not infraorbital—stimulation. The current findings may justify a rigorously masked, site-matched trial rather than establish clinical effectiveness.

Avoid comparing this study’s percentage change in mean pain score with another study’s ≥50% responder proportion because these are different estimands.

4.3 Clinical interpretation

The intervention required 24 supervised clinic sessions in four weeks, and 54 of 137 screened individuals did not proceed because they could not commit to that schedule. Feasibility and effectiveness in routine care may therefore differ from efficacy among the selected participants. Treatment burden, cost, travel, durability, and comparison with established preventive options require evaluation.

The inert sham did not reproduce sensory stimulation. Because efficacy outcomes were participant-reported, expectancy and performance effects may account for part of the observed active–sham differences. The present data do not support an acute-treatment claim.

4.4 Strengths and limitations

Strengths include a three-arm controlled design, assessment of two trigeminal sites, multidimensional outcomes, and reported complete follow-up after allocation.

The study has substantial limitations:

- 1. Ethics approval and prospective registration are not verifiable until identifiers, dates, and records are supplied.
- 2. The reported fixed block size of four is incompatible with a balanced 1:1:1 permuted-block design and requires source verification.
- 3. The original sample-size basis and any post-initiation changes to the primary outcome or analysis are unclear.
- 4. The primary analysis did not adjust for baseline and relied heavily on separate within-group tests.
- 5. The inert sham probably compromised sensory masking, and all efficacy outcomes were participant-reported.
- 6. Sham placement matched only the supraorbital site.
- 7. The study was not powered for equivalence or non-inferiority between active sites.
- 8. Exclusion of 54 screened individuals based on ability to attend restricts applicability.
- 9. Attack definition, diary windows, and derivation of non-integer monthly frequencies require clarification.
- 10. MIDAS at one month overlaps the pretreatment recall period; VAS/NRS and MPQ-5 measurement properties were incompletely specified.
- 11. Clinical baseline characteristics and concomitant-treatment data were insufficient.
- 12. Open-ended adverse-event ascertainment may have underdetected harms.
- 13. The single-center sample was small and predominantly female, with only three months of follow-up.

The previously reported analysis assigning nonrandomized individuals to sham was invalid and has been removed.

4.5 Future research

A definitive study should follow the IHS guideline for neuromodulation-device trials [45] and use a verifiable, prospectively registered protocol and SAP; concealed allocation with suitable variable block sizes; an active sensory sham matched separately to each electrode site; blinded outcome assessment; standard 28-day migraine-day outcomes [52]; validated patient-reported measures; class-specific medication rules; adequate power for the intended contrasts; systematic harms capture [49]; and longer follow-up. A factorial or double-dummy design could separate stimulation-site effects from sham-site and sensory-expectation effects.

5. Conclusions

Among a selected group able to attend 24 clinic sessions, the reported aggregate data showed fewer self-reported migraine attacks at three months with supraorbital and infraorbital TENS than with an inert sham. These findings are hypothesis-generating. Unverified allocation and registration details, likely compromised participant masking, a site-mismatched sham, outcome and analysis limitations, and inadequate power for the active-site comparison preclude firm conclusions about efficacy, safety, or equivalence. Source-document verification and prespecified participant-level reanalysis are necessary before the findings can support clinical recommendations.

Declarations

Ethics approval and consent

[Full committee], approval [identifier], approved [date]. Written informed consent was obtained from all participants.

Trial registration

IRCT [identifier], registered [date], [URL]. First enrollment: [date].

Protocol and statistical analysis plan

Available at [persistent URL]. Deviations are listed in Supplementary Table S1.

Funding

This research received [verify] no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing interests

The authors declare [insert verified declaration].

Data and code availability

De-identified participant-level data, the data dictionary, and executable analysis code will be available at [repository] subject to [specific ethical/privacy restrictions]. “Available on reasonable request” should be retained only if repository sharing is not ethically possible and the access process is defined.

Author contributions

Insert complete CRediT roles for each named author. Remove the placeholder statement.

Required supplementary material

- 1. CONSORT 2025 checklist.
- 2. CONSORT non-pharmacological-treatment checklist.
- 3. TIDieR checklist.
- 4. Dated protocol and all amendments.
- 5. Dated statistical analysis plan.
- 6. Registry history and protocol-to-publication deviation table.
- 7. CONSORT participant-flow diagram.
- 8. Randomization and envelope implementation audit.
- 9. De-identified data dictionary and reproducible code.
- 10. Intervention montage photographs/diagrams and device specifications.
- 11. Complete masking cross-tabulation.
- 12. Complete harms table.

Corrected and added references

Retain the original numbering for unchanged references during revision, correct references 11, 21, 31, and 38 as shown in the accompanying audit, and add references 41–56 below. Renumber the complete list automatically after the target journal is selected.

38. Zandifar A, Banihashemi M, Haghdoost F, Masjedi SS, Manouchehri N, Asgari F, Najafi MR, Ghorbani A, Zandifar S, Saadatnia M. Reliability and validity of the Persian HIT-6 questionnaire in migraine and tension-type headache. *Pain Pract.* 2014;14(7):625–631. doi:10.1111/papr.12120.

41. Stovner LJ, Hagen K, Linde M, Steiner TJ. The global prevalence of headache: an update, with analysis of the influences of methodological factors on prevalence estimates. *J Headache Pain.* 2022;23:34. doi:10.1186/s10194-022-01402-2.

42. Ashina M, Katsarava Z, Do TP, et al. Migraine: epidemiology and systems of care. *Lancet.* 2021;397(10283):1485–1495. doi:10.1016/S0140-6736(20)32160-7.

43. Puledda F, Silva EM, Suwanlaong K, Goadsby PJ. Migraine: from pathophysiology to treatment. *J Neurol.* 2023;270(7):3654–3666. doi:10.1007/s00415-023-11706-1.

44. Stanak M, Wolf S, Jagoš H, Zebenholzer K. The impact of external trigeminal nerve stimulator on prevention and acute treatment of episodic and chronic migraine: a systematic review. *J Neurol Sci.* 2020;412:116725. doi:10.1016/j.jns.2020.116725.

45. Tassorelli C, Diener HC, Silberstein SD, et al. Guidelines of the International Headache Society for clinical trials with neuromodulation devices for the treatment of migraine. *Cephalalgia.* 2021;41(11–12):1135–1151. doi:10.1177/03331024211010413.

46. Evans AG, Horrar AN, Ibrahim MM, et al. Outcomes of transcutaneous nerve stimulation for migraine headaches: a systematic review and meta-analysis. *J Neurol.* 2022;269(8):4021–4029. doi:10.1007/s00415-022-11059-1.

47. Yuan H, Orr SL, Al-Karagholi MAM, et al. International Headache Society evidence-based guidelines on the use of non-invasive neuromodulation devices for the acute and preventive treatment of migraine. *Cephalalgia.* 2025;45(10). doi:10.1177/03331024251388377.

48. Hopewell S, Chan AW, Collins GS, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ.* 2025;389:e081123. doi:10.1136/bmj-2024-081123.

49. Junqueira DR, Zorzela L, Golder S, et al. CONSORT Harms 2022 statement, explanation, and elaboration: updated guideline for the reporting of harms in randomised trials. *BMJ.* 2023;381:e073725. doi:10.1136/bmj-2022-073725.

50. Ashina S, Terwindt GM, Steiner TJ, et al. Medication overuse headache. *Nat Rev Dis Primers.* 2023;9:5. doi:10.1038/s41572-022-00415-0.

51. Zandifar A, Asgari F, Haghdoost F, Masjedi SS, Manouchehri N, Banihashemi M, et al. Reliability and validity of the Migraine Disability Assessment Scale among migraine and tension type headache in Iranian patients. *Biomed Res Int.* 2014;2014:978064. doi:10.1155/2014/978064.

52. Diener HC, Tassorelli C, Dodick DW, et al. Guidelines of the International Headache Society for controlled trials of preventive treatment of migraine attacks in episodic migraine in adults. *Cephalalgia.* 2020;40(10):1026–1044. doi:10.1177/0333102420941839.

53. GBD 2023 Headache Collaborators. Global, regional, and national burden of headache disorders, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet Neurol.* 2025;24(12):1005–1015. doi:10.1016/S1474-4422(25)00402-8.

54. Zhou S, Hussain N, Abd-Elsayed A, Boulos R, Hakim M, Gupta M, Weaver T. Peripheral nerve stimulation for treatment of headaches: an evidence-based review. *Biomedicines.* 2021;9(11):1588. doi:10.3390/biomedicines9111588.

55. Cocores AN, Smirnoff L, Greco G, Herrera R, Monteith TS. Update on neuromodulation for migraine and other primary headache disorders: recent advances and new indications. *Curr Pain Headache Rep.* 2025;29:47. doi:10.1007/s11916-024-01314-7.

56. Ye Y, Xie X, Huang H, Zhang H, Yu Y, Qu W. Efficacy and safety of transcutaneous supraorbital nerve stimulation versus topiramate for migraine prevention: a non-inferiority randomized trial. *J Headache Pain.* 2026;27:143. doi:10.1186/s10194-026-02367-2.

References to remove or restrict

- Reference 17 is an abstract-only pilot report and should not support an efficacy claim.

- Reference 24 concerns implanted occipital plus supraorbital stimulation and must not support infraorbital TENS.
- Reference 31 is a meeting abstract with incorrect metadata in the supplied manuscript and does not validate MPQ-5 as a continuous longitudinal endpoint.
- Reference 16 supports post-marketing tolerability and satisfaction, not efficacy.
- References 18–20 are uncontrolled exploratory studies.
- Reference 25 (TEAM) concerns acute treatment, not prevention.
- Reference 40 concerns vagus nerve stimulation and is only indirect evidence for the present intervention.